

Learning Whom to Test: Per-Case Credit Assignment for Laboratory Test Allocation in Arbovirus Surveillance

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Abstract

When dengue and chikungunya co-circulate, clinical triage cannot reliably tell them apart, and laboratory confirmation is too scarce to be applied to every suspected case. Deciding *which* suspects to test is a sequential decision made in epidemic time: tests take days to return, results change what should be done next, and hundreds of patients are open at once. We cast this problem as a Markov decision process in a simulator grounded in surveillance data—SEIR dynamics with literature parameters, spatial case distributions estimated by kriging from notifications in Rio de Janeiro and Recife, noisy clinicians, delayed laboratory results, and suspects that have neither disease. In real data, the location of a case barely reveals its disease: a Bayes classifier using location alone is right 52–57% of the time in every sizeable epidemic year of two cities (95% bootstrap intervals within ± 0.01), so tests are the main source of information. Standard deep reinforcement learning fails to learn to use them, because the value of a test is diluted across interleaved decisions about other patients (correlation between the standard return and the test’s outcome: -0.02 ; along the patient’s own timeline: 0.99). We propose a per-case variant of proximal policy optimization that computes advantages along each patient’s timeline, discounted in epidemic days, without changing the environment or the reward. Across four training geographies and four test geographies (three training seeds, 30 paired outbreaks per test), per-case credit outperforms standard advantage estimation in every one of the 16 combinations. Trained on Rio de Janeiro, the agent matches the best expert rule we could write—sequential testing guided by the clinician’s suspicion—in reward ($+27$, 95% CI $[-119, +157]$) and accuracy, with 8% fewer tests, and keeps matching it on two held-out epidemic years of another city. When location is informative (an idealized geography), the agent surpasses the rule ($+570$ [$+436, +694$]) with 63% fewer tests, but that policy does not transfer to real cities. An agent trained on a small, fixed urban footprint develops a spatial shortcut that does not transfer, a failure mode that input ablations expose.

Keywords: reinforcement learning; credit assignment; arboviruses; diagnostic testing; resource allocation; transfer; epidemiological surveillance.

1 Introduction

Arboviral diseases are among the most persistent public-health burdens in tropical countries such as Brazil [1]. Dengue, chikungunya and Zika share the *Aedes aegypti* vector and cause large cyclical epidemics that periodically overwhelm health services, with costs that range from lost productivity to hospital and emergency care [2, 3]. When these viruses co-circulate, their clinical pictures overlap, which hampers real-time surveillance [4].

The case definition used by the Brazilian Ministry of Health—sudden fever with arthralgia, myalgia or rash—is sensitive enough to capture most infections, but has low specificity for the viral etiology [5]. Distinguishing dengue from chikungunya from history and physical examination

alone is error-prone [6], and a sizeable fraction of suspects turns out to have neither disease. Laboratory confirmation resolves the ambiguity, but during fast-growing outbreaks testing every suspect is financially and logistically infeasible, so managers must ration tests. In practice this rationing is often reactive: tests are spent on clinically obvious cases while atypical or epidemiologically strategic ones remain unconfirmed [4].

Allocating laboratory tests is therefore a *sequential decision problem under uncertainty*. Testing a suspect, relying on epidemiological evidence, waiting for more information, or assigning a final diagnosis consumes finite resources and shapes the quality of future surveillance [7, 8]. Because the true state of the epidemic is only partially observed—through reporting delays, clinical noise and asymptomatic infection—good policies must reason from incomplete information and update decisions over time [8, 9]. Reinforcement learning (RL) is a natural framework for such policies, and simulation environments with a standard interface make it possible to train and compare them [10, 11].

We deliberately keep the decisions inside *epidemic time*: suspects arrive day by day, tests return after a laboratory delay, and a case comes back to the decision-maker when its result is available. This choice is what makes the problem realistic, and it is also what makes it hard to learn. With hundreds of cases open at once, the consequence of a decision about one patient arrives interleaved with the consequences of dozens of decisions about others. Standard RL estimators chain consecutive steps of the trajectory, so the value of ordering a test must travel through many unrelated decisions before it is credited—and in our setting it does not survive the trip.

Contributions.

1. **A data-grounded simulator of test allocation in epidemic time.** Dengue and chikungunya outbreaks follow SEIR dynamics with literature parameters; case locations are drawn from kriging surfaces estimated from notifications in Rio de Janeiro and Recife; suspects include non-arboviral febrile illness; clinicians are noisy; laboratory results return after five days and send the case back to the agent (Section 2).
2. **Evidence, with uncertainty, that location is weakly informative in real data.** A spatial bootstrap of the notifications shows that location predicts the disease 52–57% of the time in every sizeable epidemic year of both cities, far from the 94% of idealized generators (Section 5.1).
3. **A diagnosis of why standard deep RL fails, and a fix.** The standard return is uncorrelated with the outcome of a test; a return along the patient’s own timeline is almost perfectly correlated. Per-case PPO, which estimates advantages along each patient’s timeline in epidemic days, makes the agent investigate without changing the environment or the reward (Sections 3.3 and 5.2).
4. **A cross-geography evaluation.** Agents trained in four geographies are tested in four, with paired outbreaks and a hierarchical bootstrap over training seeds and outbreaks. Per-case credit wins everywhere; the learned policy matches the best expert rule in real cities with fewer tests, surpasses it where geography is informative, and transfers from Rio de Janeiro to held-out years of Recife—but not from idealized to real geography, nor from a small fixed urban footprint to a larger one (Sections 5.3 and 5.4).

2 Simulation environment

The environment implements the Gymnasium interface [10, 12]. Each episode simulates 300 days of concurrent dengue and chikungunya outbreaks in one city. Every parameter reported below

is set in version-controlled configuration files, and the environment’s behavior is locked by an automated test suite.

2.1 Epidemic dynamics

Each disease follows a frequency-dependent SEIR model in a closed population [13, 14]:

$$\dot{S} = -\beta \frac{SI}{N}, \quad \dot{E} = \beta \frac{SI}{N} - \sigma E, \quad \dot{I} = \sigma E - \gamma I, \quad \dot{R} = \gamma I, \quad (1)$$

with $\sigma = 1/\text{latency}$, $\gamma = 1/\text{infectious period}$ and $\beta = R_0\gamma$. Notifications follow the flow σE (onset of the infectious phase, which we take as symptom onset); the cumulative curve is integrated numerically and daily new cases are its rounded increments.

Because both viruses pass through the mosquito, the interval between successive cases includes the extrinsic incubation. Rather than a full host–vector model, we use an *effective* human latent period that absorbs both incubations, so that the model’s generation time (latency plus infectious period) matches the serial interval observed in the field (Table 1). For dengue, the intrinsic incubation averages 5.9 days [15], human infectiousness lasts about five days after symptom onset [16], and space–time clustering is strongest at 15–17 days [17]; R_0 in Rio de Janeiro was estimated at 1.70 (2002) and 1.25 (2012) [18]. For chikungunya in Rio, $R_0 = 1.56$ (95% CI 1.46–1.67) assuming a 14-day generation time [19]; only the sum of latency and infectious period is anchored by data. Each episode draws R_0 uniformly within these ranges, with a population of 300 per disease and 1% initially infected. We validate the generator against properties that can be checked against theory—the growth rate implied by R_0 and the final-size equation $z = 1 - e^{-R_0 z}$ (Figure 1)—rather than against the shape of the curves, which can look plausible when the epidemiology is wrong.

Disease	Effective latency	Infectious period	Generation time	R_0
Dengue	11 d	5 d	16 d	1.25–1.70
Chikungunya	8 d	6 d	14 d	1.46–1.67

Table 1: SEIR parameters. R_0 is drawn uniformly within the range at each episode. Sources: [15–19].

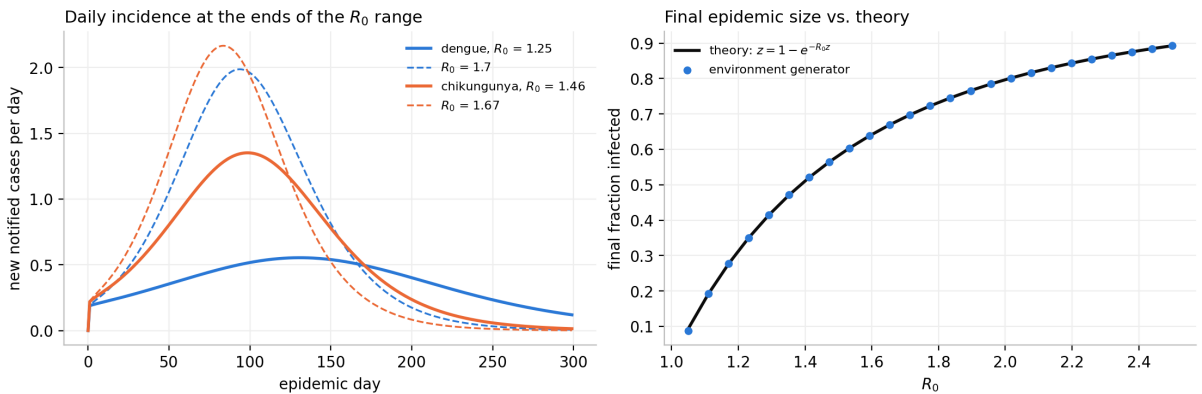


Figure 1: Epidemic dynamics. Left: daily incidence at the ends of the R_0 range for each disease; epidemics last several months. Right: final fraction infected produced by the generator against the final-size equation.

2.2 Where cases occur: spatial surfaces from surveillance data

Cases are placed on a 400×400 grid by an interchangeable spatial generator. The default generator uses surfaces estimated by ordinary kriging from georeferenced notifications: counts are aggregated in 500 m cells, the log-counts are interpolated, and the result is normalized into a spatial probability per disease, $P(\text{cell} \mid \text{disease})$, from which each day’s cases are sampled. At every episode one of four rigid transformations (identity, 180° rotation, horizontal or vertical mirror) is applied *jointly* to both surfaces, so the agent cannot memorize one city’s geography while the relative geometry of the two diseases—and hence the difficulty—is preserved.

Rio de Janeiro. The reference surfaces use dengue and chikungunya notifications of Rio de Janeiro in 2015–2016 with residence coordinates. Chikungunya barely circulated in Rio in 2015 (70 notifications, against about 14 thousand in 2016), so alternative Rio scenarios combine dengue from 2015 or 2016 with chikungunya from 2016.

Recife. As a second city we use the open notification microdata of Recife [20], which cover dengue (2013–2025) and chikungunya (2015–2025). Each record carries the final classification (including discarded cases), the confirmation criterion, laboratory results and the residence street and neighborhood, but no coordinates. We geocode each record against the city’s official neighborhood polygons and street segments [21]: the street name is matched (after normalizing abbreviations) to the streets that cross the reported neighborhood, or a neighboring one, and a point is drawn along the matched street; when the street is not found, a point is drawn uniformly within the neighborhood. Of 144,175 confirmed or discarded notifications of Recife residents, 72% were placed on the reported street within the reported neighborhood, 6% on the same street in an adjacent neighborhood, 22% at neighborhood level, and fewer than 1% could not be placed. We build one surface per year (2015–2025) from cases confirmed by any criterion, and set the probability to zero outside the municipality, which also removes the sea.

Spatial bootstrap. To quantify the uncertainty of the surfaces and to give training more spatial variety, we build 20 bootstrap replicates for each city and year: the notifications of each disease are resampled with replacement and the kriging is recomputed; in Recife, each replicate is also re-geocoded with a different seed, so the uncertainty of street- and neighborhood-level placement is included. Chikungunya notifications of 2015–2016 coded with the legacy dengue classification scale (4,888 records without any chikungunya-specific result) are excluded as ambiguous.

How informative is location? What matters for the decision problem is not whether each disease has sharp foci, but whether the two diseases occupy *different* places. We summarize this with the accuracy of the Bayes classifier that sees only the location of a case, with equal priors: $\frac{1}{2} \sum_{\text{cells}} \max\{P(\text{cell} \mid \text{dengue}), P(\text{cell} \mid \text{chik})\}$. It equals 0.5 when location carries no information and 1 when it determines the disease. In Rio 2015–2016 it is 0.544: each surface has clear foci, but the foci coincide (correlation 0.85 between the two surfaces; in 90% of the cells one disease is at most 1.4 times more likely than the other), because both follow where people live and where the vector is (Figure 2, top). The idealized generator used in earlier work—two truncated Gaussian foci—reaches 0.937.

Controlling spatial informativeness. To study sensitivity to this quantity we sharpen the surfaces with a temperature τ ($p \propto p^\tau$; $\tau = 0$ is uniform), which moves the Bayes accuracy from 0.50 ($\tau = 0$) through 0.544 ($\tau = 1$) to 0.94 ($\tau = 32$). Sharpening reaches high separability only by concentrating each disease: half of the dengue cases fall in 34% of the cells at $\tau = 1$, in 1% at $\tau = 8$ and in 8 of 13,195 cells at $\tau = 32$ (Figure 2, bottom). Intermediate values preserve the shape of the foci; $\tau \geq 16$ is an artificial limit rather than a plausible epidemic.

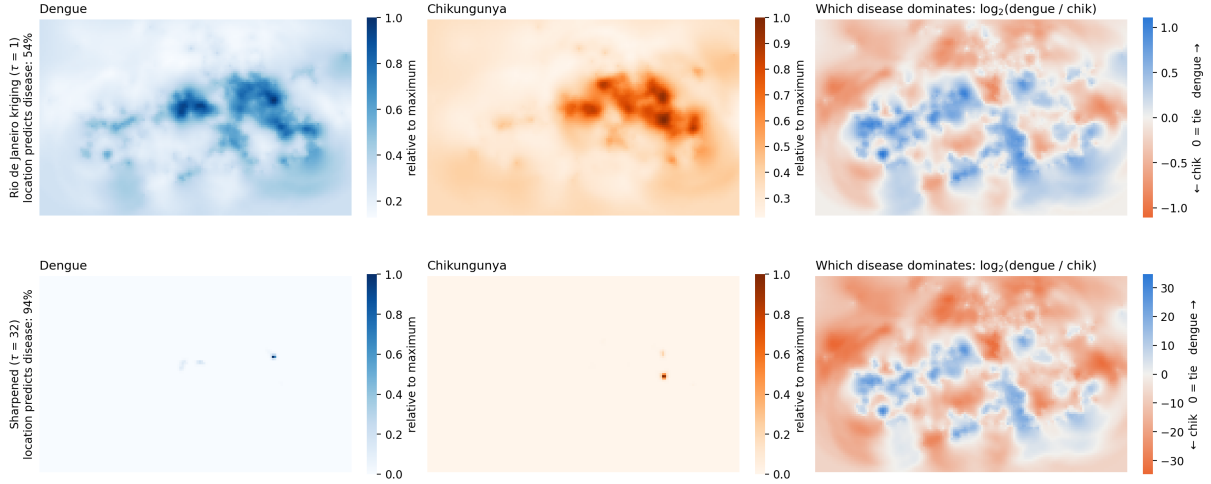


Figure 2: Spatial surfaces of dengue and chikungunya, each on its own scale, and the log-ratio between them. Top: kriging on Rio de Janeiro notifications (2015–2016); each disease has sharp foci, but the ratio stays between $0.5\times$ and $2\times$ and location predicts the disease 54% of the time. Bottom: sharpened surfaces ($\tau = 32$); location becomes highly informative only because each disease collapses onto a handful of cells.

2.3 Case mix: suspects that are neither disease

In real triage, many suspects have another febrile illness. The simulator adds non-arboviral suspects at 25% of the notified cases, uniform in space (they do not follow the vector’s foci) and proportional in time to the arboviral cases of each day. This proportion is consistent with the Recife data: in the epidemic years (2015, 2016 and 2021), discarded notifications are 18–30% of the total. These cases give depth to the investigation: a negative dengue test no longer implies chikungunya, a second test becomes informative, and concluding “not an arbovirus” becomes a legitimate decision.

Two-disease variant. An earlier version of our kriging generator omitted the non-arboviral suspects, so every suspect had one of the two diseases. We keep this configuration as an ablation of the case mix—the *two-disease variant*—because it isolates how the case mix changes which policy is optimal (Section 5.3). Several robustness analyses were run in this variant and are reported as such.

2.4 Clinicians

The true disease is hidden from the agent and replaced by a noisy clinical label. Each episode draws a clinician whose accuracy is uniform in $[0.50, 0.95]$ and applies to both diseases. Non-arboviral suspects are recognized as such at triage with probability 0.15; otherwise the clinician labels them dengue or chikungunya with equal probability—if they were easy to recognize, they would not be in the queue of suspects. The environment also supports Beta-distributed sensitivity and specificity with controllable mean and dispersion. The clinician turns out to be the dominant source of variance between episodes (Section 4).

2.5 Laboratory tests and epidemiological confirmation

Laboratory tests follow RT-PCR as the diagnostic gold standard: sensitivity 0.95, specificity 0.98 and 2% indeterminate results, which leave the working diagnosis unchanged. **[TODO: anchor the RT-PCR parameters in references; they vary with the kit, the day of collection]**

and the serotype.] Results become available five days after collection. Epidemiological confirmation simulates a field investigation: a case is confirmed if, within a radius of 20 grid cells, there is more than one other case already confirmed *by tests the agent ordered*. It therefore uses only information the surveillance system has, and it makes each positive test an investment in future confirmations nearby. Under a policy that tests every case, it confirms 19.3% of cases with 80.1% precision, against 57.1% accuracy of the clinical label.

2.6 Decision process

The epidemic advances in days; within each day, the cases awaiting a decision are presented to the agent one at a time, so each agent step is a decision about one case (Figure 3). The agent observes a six-channel spatial tensor summarizing notified cases, tests and confirmed cases; the coordinates of the current case; and a 16-dimensional context vector with the clinical label, the results of tests already done, epidemiological confirmation, the local density of confirmed cases, and a two-number summary of the evidence accumulated about the clinician (agreement between test results and clinical labels so far, and how much evidence supports it). This summary tracks the clinician’s true accuracy with correlation 0.99; estimating it requires spending some tests early, which creates a natural exploration–exploitation trade-off.

Actions. For each case the agent chooses among seven actions: **no action**, which costs nothing and lets the clinician’s diagnosis stand; **investigate**, by ordering a dengue test, a chikungunya test, or epidemiological confirmation; and **conclude**, by asserting dengue, chikungunya or “other”. The asserted class comes from the action itself, so the agent can disagree with the clinician without ordering a test.

Case lifecycle. When the agent investigates, the sample is collected immediately but the result matures five days later; **when the result arrives, the case returns to the agent**, who decides with the result in hand. A case leaves the queue when it is concluded, when the agent takes no action on it, or when the episode ends; it can return at most twice, after which only concluding actions are allowed. Without returns, ordering a test would mean giving up the decision about that case, and testing and deciding would be mutually exclusive.

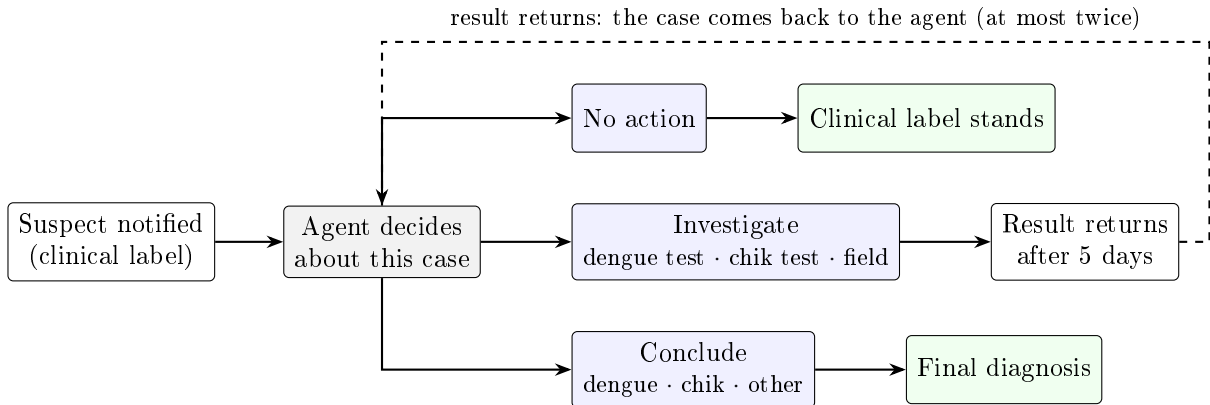


Figure 3: Lifecycle of a suspected case. Each agent step is a decision about one case; hundreds of cases are open at the same time, so the steps of one case are interleaved with those of many others.

2.7 Reward

The reward has three layers. **Immediate cost:** each test or field investigation costs 4.0; no action and concluding cost nothing. **Decision outcome:** a concluding action earns +10 if correct, −30 if it discards a real arbovirus case (a surveillance false negative), and −20 for other errors; concluding an investigated case earns a +2 bonus. **Terminal score:** at the end of the episode each case contributes +1 if its final diagnosis is correct and −3 otherwise, whether or not it was tested, so a test is rewarded for the information it brings and not for being ordered. A case whose investigation was opened and never concluded costs −10, charged when it is abandoned; taking no action on an uninvestigated case is not abandonment but acceptance of the clinical diagnosis, judged by the terminal score.

The test cost governs whether an allocation problem exists: with cheap tests, testing everyone is optimal; with expensive ones, the best policy is to do nothing. At 4.0, investigating pays with a narrow margin—the regime in which choosing *which* cases to investigate matters most. **[TODO: repeat the test-cost sweep in the full environment; the current calibration was done with the idealized generator.]**

Per-case decomposition. At each step, the environment also reports which case the step’s reward belongs to, on which epidemic day the decision was made, and whether the case is finished; each case’s terminal share is attributed to it exactly once. The per-case shares sum exactly to the episode reward, so the metric used to compare agents is unchanged. This decomposition is the raw material for the credit assignment of Section 3.3.

3 Agents

3.1 Fixed policies

Fixed policies establish what can be achieved without learning; without them, a trained agent can look competent merely by beating chance.

- **Clinical only:** never acts; the clinician’s label stands.
- **Confirm-all:** concludes every case with the clinical label, without testing.
- **Test-once:** orders one dengue test per case and concludes with the result.
- **Test-twice:** orders both tests for every case and concludes with both results.
- **Sequential, dengue first:** orders a dengue test; orders the chikungunya test only if the first is not positive; concludes.
- **Sequential, clinician-guided:** as above, but first tests for the disease the clinician suspects (dengue if the clinician suspects neither).

The sequential rules are the natural expert policies between test-once and test-twice: a positive result settles the case, a negative or indeterminate one calls for the other test.

3.2 Learning agents

Deep Q-network. We first trained a DQN [22] with a convolutional encoder. It is kept as a control: once stabilized with a Huber loss, it converges reliably to a degenerate policy of inaction (Section 5.2).

Proximal policy optimization. PPO [23] optimizes a stochastic policy π_θ with the clipped objective

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_t[\min(\rho_t(\theta)\hat{A}_t, \text{clip}(\rho_t(\theta), 1 - \epsilon, 1 + \epsilon)\hat{A}_t)], \quad \rho_t(\theta) = \frac{\pi_\theta(a_t | s_t)}{\pi_{\theta_{\text{old}}}(a_t | s_t)}, \quad (2)$$

where the advantage $\hat{A}_t$ is usually obtained by generalized advantage estimation (GAE) [24] over *consecutive* steps of the trajectory:

$$\delta_t = r_t + \gamma V(s_{t+1}) - V(s_t), \quad \hat{A}_t = \sum_{k \geq 0} (\gamma \lambda)^k \delta_{t+k}. \quad (3)$$

3.3 Per-case credit assignment

The problem. Consecutive steps of the trajectory belong to different patients. Between a test and the decision it informs there are about 30 decisions about *other* cases; the test itself only returns -4 , and its benefit appears in a later step that belongs to the same case but is surrounded by unrelated rewards. Equation (3) forces that value through all of them, accumulating their noise. We measured the effect directly: at the step of each test, the correlation between the return and whether the test led to a correct diagnosis is -0.019 for the standard return discounted per step, and 0.994 for the return of the case itself discounted per epidemic day. The signal is in the trajectory; the standard definition of the return erases it.

This also explains why three earlier attempts failed. Potential-based reward shaping [25] does not apply: the next state is already another patient, so the potential cancels within the step, and with hundreds of open investigations the drift $(1 - \gamma)\Phi$ exceeds the intended increment for any scale. Retroactive credit would require rewriting rewards already delivered. Making tests cheaper doubled the margin of investigating and did not make the agent order a single test. The obvious alternative—one episode per case—would remove the interleaving by destroying the epidemic dynamics, which is the object of this work.

The method. Each transition carries the episode and case identifiers, the epidemic day d of the decision and the case’s share r^{case} of the reward. *Per-case PPO* groups the transitions of a batch by (episode, case) and computes GAE along each patient’s timeline, with discounting raised to the number of *days* between decisions rather than steps. For the successive decisions j of one case:

$$\begin{aligned} \text{disc}_j &= \gamma^{d_{j+1} - d_j}, \\ \delta_j &= r_j^{\text{case}} + \text{disc}_j V(s_{j+1}) - V(s_j), \\ \hat{A}_j &= \delta_j + \text{disc}_j \lambda \hat{A}_{j+1}. \end{aligned} \quad (4)$$

At the last step of a case in the batch there is no future if the case is finished; if the batch cut its timeline, we bootstrap from its own value with one day of discount. The structure mirrors credit assignment in multi-agent systems, with each case as an agent sharing the world with the others. **[TODO: cite multi-agent credit-assignment literature.]** Two invariances are verified by automated tests: the per-case shares reconstruct the episode reward, so the evaluation metric is unchanged; and the advantage of a case does not depend on the interleaving—inserting 30 noisy decisions about other cases between two steps of case X leaves X ’s advantage unchanged.

Limitation. Per-case credit ignores the externality of a test: a positive result improves epidemiological confirmation for *future* cases nearby, and that benefit does not flow back to the case that paid for the test. The agents barely use epidemiological confirmation, so the effect is currently small.

3.4 Architecture and training

All deep agents share the same network: convolutional layers process the spatial tensor and end in adaptive pooling to $64 \times 6 \times 6$, and a dense branch processes the case coordinates and context vector [26]. Since the pooling discards resolution, the map is delivered at 100×100 , which reduced observations sixteen-fold at no measurable cost. We use Tianshou [27] with 300,000 steps per run, learning rate 3×10^{-4} , $\gamma = 0.99$, $\lambda = 0.95$, $\epsilon = 0.2$, entropy coefficient 0.02, batches of 256 and eight parallel training environments. The action mask that forces a conclusion after two returns is applied to the actor’s logits—otherwise PPO silently ignores it. Rewards are scaled by 0.10 during training only.

To separate the effect of credit assignment from that of temporal information, we trained three arms with the same network and hyperparameters (Table 2): A $\times$ C isolates credit assignment with identical observations; B $\times$ C isolates four optional temporal features (fraction of the horizon elapsed, cases notified today, 7-day trend, age of the case), all computed from what surveillance observes.

Arm	Advantage	Context	Training seeds
A	standard GAE (Eq. 3)	16	4 (two-disease); 3 per geography (full)
B	per case, in days (Eq. 4)	16 + 4 temporal	3 (two-disease)
C	per case, in days (Eq. 4)	16 (as A)	4 (two-disease); 3 per geography (full)

Table 2: PPO arms.

4 Evaluation protocol

Paired outbreaks. Agents are evaluated on 10 test outbreaks, each identified by a fixed seed that reproduces the same epidemic, the same clinician and the same clinical noise for every agent, so comparisons are *paired*. This matters: the clinician drawn in an episode explains most of the variance—its accuracy correlates at 0.97 with the episode reward of both the clinical policy and a trained agent. A second set of 12 never-used seeds tests generalization. We always evaluate the *final* checkpoint; selecting the best checkpoint during training selects noise (it scored 1,000–1,460 points above the final checkpoint for DQN).

Hierarchical, paired bootstrap. There are two sources of variance: the seed that trained the agent and the outbreak it is evaluated on. Confidence intervals come from a hierarchical bootstrap [28] that resamples training seeds and then outbreaks; outbreak resampling is *shared* across arms within each replicate, preserving the pairing, while training seeds—*independent* runs—are resampled separately per arm. Following Agarwal et al. [29], we report the mean, the interquartile mean, the paired difference between arms and the probability of improvement $P(X > Y)$, the chance that an episode of X beats the same episode of Y over all pairs of training seeds (ties count 1/2), with 95% percentile intervals from 10,000 replicates. With three or four training seeds the upper level has little resolution, so intervals are honest about outbreaks and optimistic about training seeds.

Metrics. The primary metric is the episode reward; secondary metrics are three-class diagnostic accuracy (dengue, chikungunya, other) and the number of laboratory tests per episode. In the two-disease variant no case is “other”, so accuracy is effectively binary.

Training geographies and cross-geography evaluation. All cross-geography experiments use the full environment (with non-arboviral suspects) and differ only in where cases occur. Arms A and C are trained with three seeds each (45–47) in four geographies:

- **Rio de Janeiro:** each episode draws one of the 20 bootstrap replicates of the 2015–2016 surface.
- **Recife:** each episode draws one replicate of one training year (2017–2020 and 2022–2025; 160 surfaces). The two largest epidemics with both diseases, 2016 and 2021, are held out for testing; 2015 is excluded (only 201 confirmed chikungunya cases, and the epicenter of the Zika epidemic).
- **Idealized:** two well-separated Gaussian foci (location predicts the disease 94% of the time).
- **Mixture:** one third of the episodes from each of the three above.

Every trained agent and every fixed policy is then evaluated in four test geographies, always on the *original* surfaces and never on training replicates: Rio de Janeiro 2015–2016, Recife 2016 and Recife 2021 (held out from Recife training), and the idealized generator. Each test uses 30 paired outbreaks with seeds never used before (3001–3030). Two further groups of per-case agents (three seeds each) test specific remedies: the mixture with twice the training budget (600 thousand steps), and Recife with the city randomly rotated, rescaled (35–100% of the grid) and repositioned at every episode.

5 Results

5.1 In real data, location barely reveals the disease

Recife offers eleven years of both diseases in one city, with epidemics of very different composition—dengue-dominated in 2015–2016 and 2019, chikungunya-dominated in 2021 (Figure 4, left). In every year, the Bayes accuracy of location alone stays close to Rio’s 0.544: between 0.520 and 0.573 from 2016 to 2025 (Figure 4, right). The highest value, 0.605 in 2015, comes from only 201 confirmed chikungunya cases, which makes that surface artificially concentrated. The places where one disease dominates change from year to year (Figure 5), but the magnitude of the difference stays small. Recife 2016, the largest epidemic with both diseases, is strikingly similar to Rio 2015–2016 (Bayes accuracy 0.557 vs. 0.544; correlation between the two diseases’ surfaces 0.81 vs. 0.85).

The spatial bootstrap confirms that these values are precise: the 95% intervals are 0.541 [0.539, 0.545] for Rio de Janeiro 2015–2016, 0.556 [0.552, 0.559] for Recife 2016 and 0.530 [0.528, 0.535] for Recife 2021, and, apart from 2015 (0.60, driven by only 201 chikungunya cases), no Recife year exceeds 0.57 (all intervals within ± 0.01). The correlation between the two surfaces, by contrast, is unstable in years with few cases (e.g., $[-0.06, 0.23]$ in 2024) and is only interpreted for the large epidemics (Rio 2015–2016: 0.80 [0.77, 0.82]; Recife 2016: 0.80 [0.79, 0.82]; Recife 2021: 0.95 [0.93, 0.96]).

The implication for the decision problem is direct. The idealized generator, with location 94% informative, lets an agent diagnose from the map; real surveillance data do not. In earlier work with that generator, a PPO agent reached a positive reward without ordering a single test, by disagreeing with the clinician based on location. In real geography, the laboratory test is the main source of information, and the allocation problem is real.

5.2 Standard deep RL does not learn to investigate; per-case credit does

DQN and standard PPO. DQN was unstable, with rewards oscillating by up to 6,300 points between neighboring evaluations. A Huber loss stabilized it unambiguously (from ± 1340 to ± 9 across seeds) and made it *worse*: stabilized, it converged reliably to 66.7% inaction and no tests (-4164 ± 9). With the same environment, network and action mask, PPO with standard

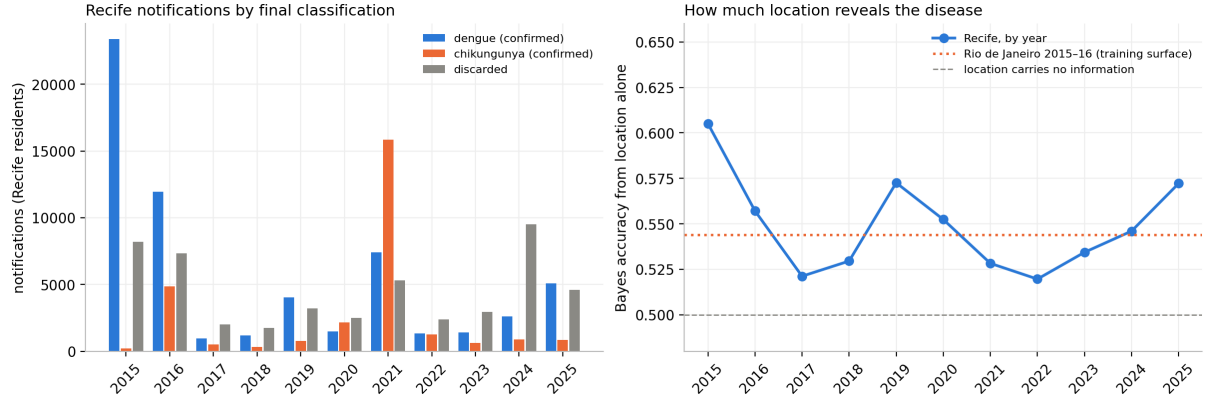


Figure 4: Recife, 2015–2025. Left: notifications of residents by final classification. Right: accuracy of a Bayes classifier that sees only the location of a case, per year, against the Rio de Janeiro surface used for training (dotted) and the no-information level (dashed).

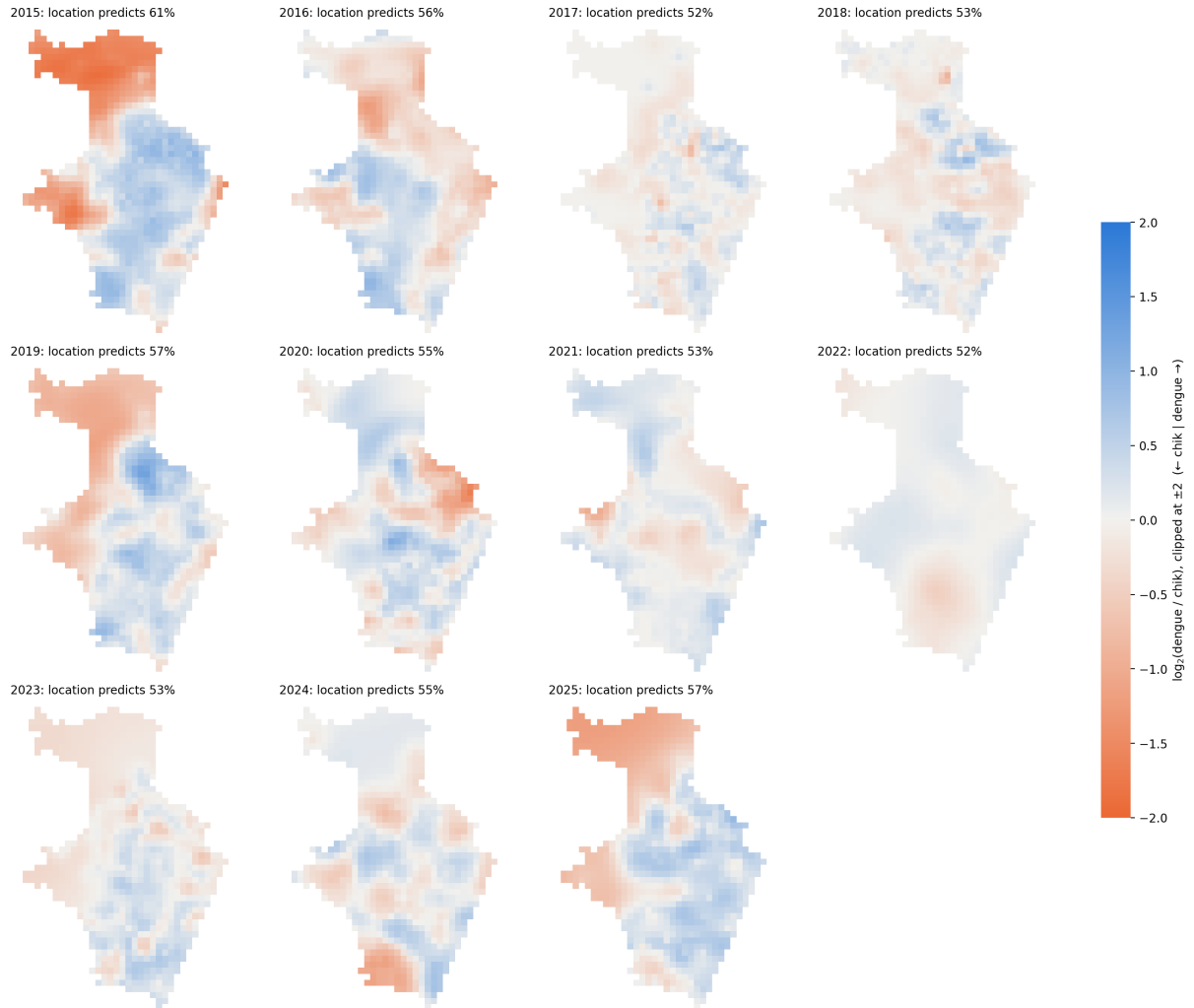


Figure 5: Where each disease dominates in Recife, year by year: $\log_2(P(\text{cell} | \text{dengue})/P(\text{cell} | \text{chik}))$ from kriging surfaces, clipped at ± 2 . The map changes between years, but the contrast remains weak.

GAE was the first learned agent to beat doing nothing ($+632 \pm 43$), with monotonic training curves. Learning rate, buffer size, n -step returns and target-update frequency did not move DQN measurably. PPO, however, hardly investigated: 7 to 37 tests per episode in the Rio geography, against about 370 for test-once.

Per-case credit. Table 3 and Figure 6 show the three arms in the two-disease variant. Three conclusions follow.

The gain comes from credit assignment. With identical observations, C outperforms A by $+2009$ [$+1197$, $+2882$] with $P(C > A) = 1.00$: in every outbreak, for every pair of training seeds, the per-case agent wins. It moves from 25 to 284 tests per episode and gains 20 accuracy points. A also collapses in specific outbreaks (down to -2200 in those with poor clinicians), where it inherits the clinician’s errors because it barely tests; C never collapses.

The temporal features do not contribute. B and C are indistinguishable (-43 [-196 , $+85$], $P = 0.47$). This does not show that temporal information is useless, only that these four features added nothing on top of per-case credit.

The per-case agent matches the best fixed rule with fewer tests. Without non-arboviral suspects, a single test settles a case and test-once is the best fixed policy. C’s reward is statistically indistinguishable from it (-90 [-250 , $+67$]) with 84 fewer tests per episode (-23%) and 2.2 fewer accuracy points. On 12 unseen outbreaks the pattern holds: $C \times A = +1653$ [$+1072$, $+2313$], $P = 1.00$; $C \times \text{test-once} = +6$ [-124 , $+149$]. An input ablation without retraining shows that zeroing the entire spatial map changes only 0–2% of the agent’s decisions: the policy decides from the case context, consistent with the weak spatial signal of Section 5.1.

Policy	Seeds	Reward [95% CI]	Accuracy [95% CI]	Tests [95% CI]
Test-once (best fixed rule)	–	$+2782$ [$+2668$, $+2900$]	0.960 [0.953, 0.966]	369 [353, 384]
C: per-case credit	4	$+2692$ [$+2523$, $+2842$]	0.937 [0.923, 0.950]	284 [223, 335]
B: per-case credit + time	3	$+2649$ [$+2413$, $+2842$]	0.931 [0.908, 0.946]	266 [206, 321]
Test-twice	–	$+1051$ [$+969$, $+1135$]	0.950 [0.945, 0.954]	737 [706, 768]
A: standard GAE	4	$+684$ [-146 , $+1498$]	0.739 [0.674, 0.803]	25 [5, 64]
Confirm-all	–	$+346$ [-656 , $+1335$]	0.710 [0.633, 0.789]	0
Clinical only	–	-64 [-184 , $+54$]	0.710 [0.633, 0.789]	0

Table 3: Two-disease variant (Rio de Janeiro surfaces), 10 paired outbreaks; final checkpoints; hierarchical, paired bootstrap. Paired contrasts: $C \times A = +2009$ [$+1197$, $+2882$], $P = 1.00$; $B \times C = -43$ [-196 , $+85$], $P = 0.47$; $C \times \text{test-once} = -90$ [-250 , $+67$], $P = 0.38$.

5.3 With non-arboviral suspects: the case mix decides the best rule

Adding non-arboviral suspects changes the problem, not just the numbers (Table 4). With only two diseases, a negative dengue test implies chikungunya and test-once is optimal. With a quarter of the suspects being neither, a negative test is ambiguous: test-once drops from $+2782$ to $+322$ and 0.756 accuracy, and investigation must go deeper. The idealized generator, which always had non-arboviral suspects, shows the same ordering as the full environment—the reordering is caused by the case mix, not by geography. The best fixed rule is now sequential testing guided by the clinician: testing first for the suspected disease settles most cases with one test when the clinician is right.

The per-case agent matches the best expert rule with fewer tests. Table 5 and Figure 7 show agents trained on Rio de Janeiro, evaluated on the original Rio surface over 30 new paired outbreaks with three training seeds. C reaches $+2175$ against $+2149$ for the clinician-guided sequential rule: the difference, $+27$ [-119 , $+157$], is not distinguishable from zero ($P(C >$

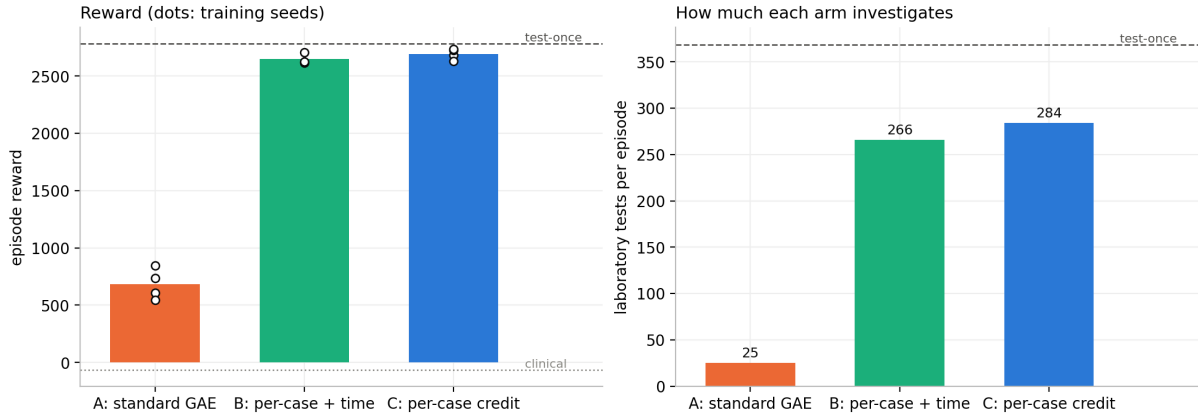


Figure 6: The three PPO arms in the two-disease variant. Left: mean episode reward (dots are training seeds). Right: laboratory tests per episode. Dashed lines: test-once; dotted: clinical only.

Fixed policy	Kriging, two-disease	Kriging, full	Idealized generator
Test-once	+ 2782	+322	+473
Test-twice	+1051	+1225	+1249
Sequential, clinician-guided	–	+ 2330	–
Confirm-all	+346	–1734	–1821
Clinical only	–64	–344	–355

Table 4: Fixed policies with SEIR dynamics on the 10 benchmark outbreaks. Removing non-arboviral suspects (two-disease variant) is what makes a single test sufficient; the idealized generator, which has them, orders the policies as the full environment does.

rule) = 0.54), and neither is the difference in accuracy (-0.007 [-0.018 , $+0.002$]). C orders 53 fewer tests per episode [-71 , -39], an 8% saving. Against the other fixed rules C is clearly better: $+376$ over sequential dengue-first and $+1031$ over test-twice, with fewer tests than both. The agent was never told about test order or the clinician’s reliability; it learned a policy as good as the best rule we could design, and a more frugal one.

The standard agent collapses. With standard GAE, A orders about 77 tests per episode, its accuracy stays near the clinician’s, and its reward falls well below doing nothing (-1604 against -368). The gap between the arms is $+3780$ [$+3217$, $+4386$], with $P(C > A) = 1.00$.

Policy	Seeds	Reward [95% CI]	Accuracy [95% CI]	Tests [95% CI]
C: per-case credit	3	+2175 [$+2010$, $+2323$]	0.934 [0.923, 0.942]	647 [615, 677]
Sequential, clinician-guided	–	+2149 [$+2043$, $+2254$]	0.941 [0.938, 0.945]	699 [669, 728]
Sequential, dengue first	–	+1799 [$+1700$, $+1897$]	0.937 [0.933, 0.941]	772 [749, 794]
Test-twice	–	+1144 [$+1064$, $+1228$]	0.937 [0.933, 0.941]	931 [899, 963]
Test-once	–	+407 [$+323$, $+488$]	0.763 [0.757, 0.768]	466 [449, 481]
Clinical only	–	-368 [-436 , -299]	0.551 [0.515, 0.588]	0
A: standard GAE	3	-1604 [-2221 , -1010]	0.590 [0.548, 0.631]	77 [20, 129]

Table 5: Full environment, agents trained and evaluated on Rio de Janeiro: 30 paired outbreaks never seen before, three training seeds per arm, hierarchical and paired bootstrap. Paired contrasts: $C \times$ clinician-guided sequential = $+27$ [-119 , $+157$], $P = 0.54$, tests -53 [-71 , -39]; $C \times A = +3780$ [$+3217$, $+4386$], $P = 1.00$.

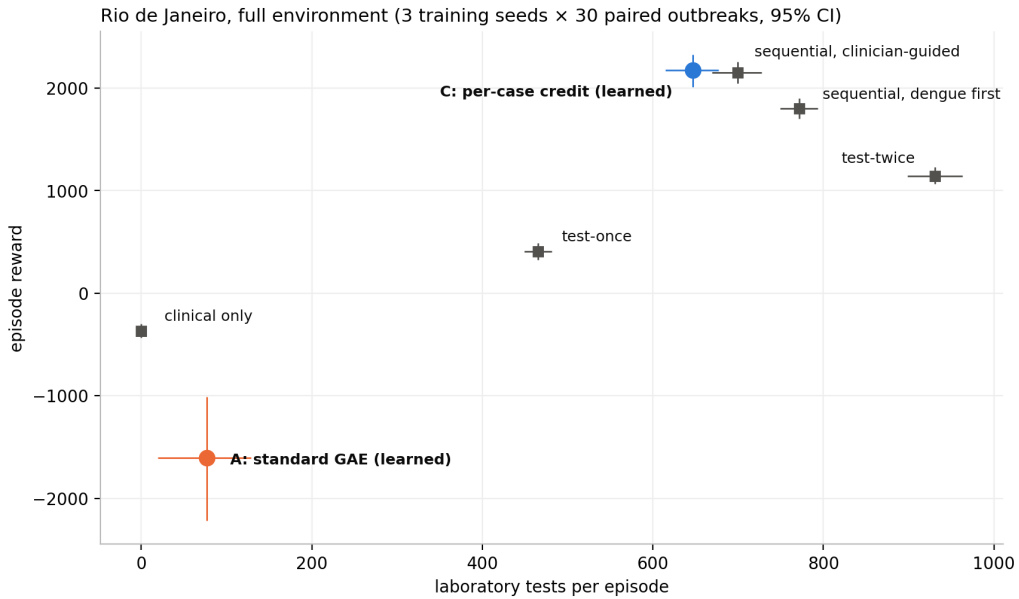


Figure 7: Reward against tests per episode in the full environment on Rio de Janeiro (means with 95% intervals over three training seeds and 30 paired outbreaks). The per-case agent sits next to the best expert rule, with fewer tests; the standard agent collapses near inaction.

5.4 Transfer across geographies

Figure 8 summarizes the cross-geography evaluation: for each training geography and each test geography, the paired difference between the per-case agent and the clinician-guided sequential rule. Four findings emerge.

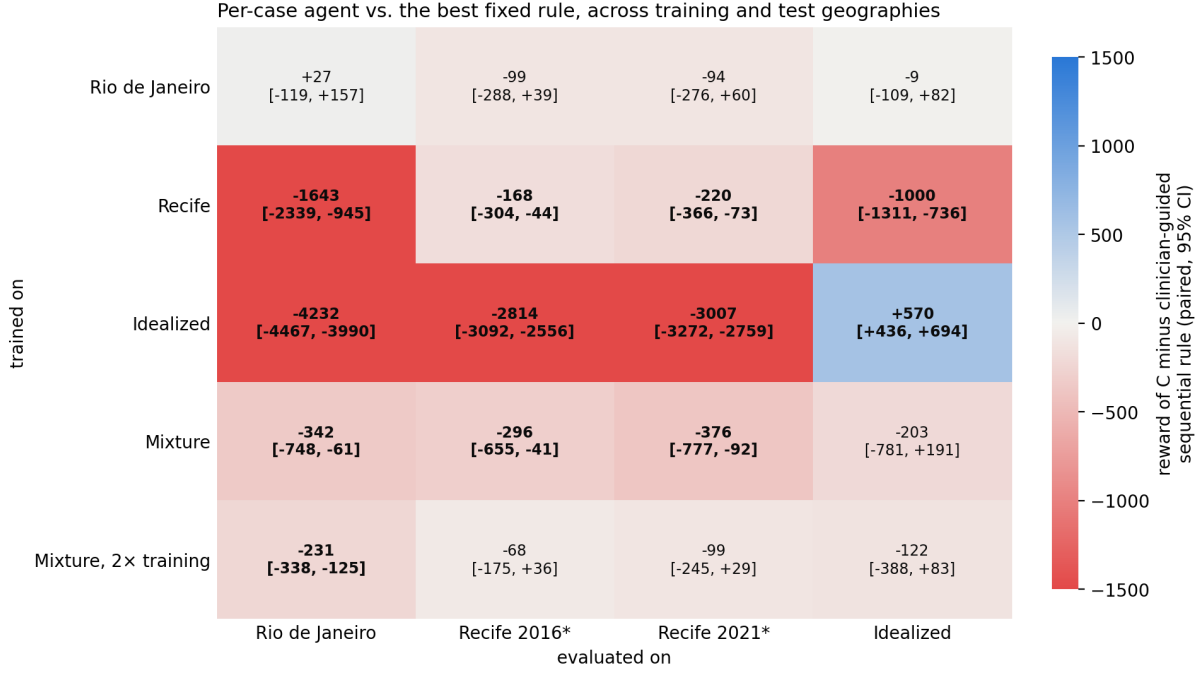


Figure 8: Per-case agent minus the clinician-guided sequential rule, by training geography (rows) and test geography (columns). Cells show the paired difference in episode reward with its 95% interval (three training seeds $\times$ 30 paired outbreaks); bold marks intervals that exclude zero. Recife 2016 and 2021 are held out from Recife training. The rule does not use location, so it is the same yardstick in every column (about +2150).

Per-case credit wins everywhere. In all 16 combinations of training and test geography, C outperforms A trained in the same geography, by +1051 to +4307 in reward, with $P(C > A)$ between 0.94 and 1.00 (1.00 in 13 of the 16). The failure of standard advantage estimation is not specific to one city or geography.

A policy learned in Rio de Janeiro transfers to another city. The Rio-trained agent stays statistically tied with the rule on the two held-out Recife years (-99 $[-288, +39]$ and -94 $[-276, +60]$), ordering 68–70 fewer tests with 1.7–1.8 fewer accuracy points, and on the idealized geography (-9 $[-109, +82]$). It is the most robust learned policy in the study.

When location is informative, learning beats the rule—but does not transfer. Trained on the idealized geography, the agent surpasses the rule on that geography by +570 $[+436, +694]$ while ordering 442 fewer tests (63% fewer), at a cost of 4.6 accuracy points: it learns to read the disease from the map and saves tests where the map is decisive. The same agent collapses in real cities (-2814 to -4232 below the rule), where location carries almost no information.

A small, fixed urban footprint induces a spatial shortcut. The Recife-trained agent falls below the rule even on its own held-out years (-168 $[-304, -44]$ and -220 $[-366, -73]$) and fails in Rio de Janeiro (-1643 $[-2339, -945]$, with large variation between seeds). Input ablations without retraining (10 outbreaks per condition) explain why. The Rio-trained agent is insensitive to its spatial inputs: zeroing the map, the case coordinates or the local density of confirmed cases leaves its reward between +2075 and +2295 in both cities. The Recife-trained agent depends on them: without the case coordinates it collapses even in Recife ($+2135 \rightarrow -4631$), and without the map it transfers much *better* to Rio ($+464 \rightarrow +1514$). Rescaling the density features to the training range does not help (+291), so the dependence is not on their scale. The Recife

surfaces occupy about half of the grid inside the municipal boundary, always at the same place up to mirroring, whereas the Rio surface covers the whole grid; the network appears to use the footprint as a cue that does not exist elsewhere. Our attempted remedy—randomly rotating, rescaling and repositioning the city at every episode—made matters worse (Recife 2016: -430 below the rule; Rio: -3491), plausibly because shrinking the city concentrates the cases and shifts every spatial input further from what other cities produce. We report this as an open problem.

Mixing geographies needs more training. With the standard budget of 300 thousand steps, the agent trained on the mixture falls below the rule in every real test (-296 to -376) and varies strongly between seeds (standard deviation 324–545). Doubling the budget stabilizes it (standard deviation 64–232) and brings it to a statistical tie with the rule on both Recife years (-68 [-175 , $+36$] and -99 [-245 , $+29$]) and on the idealized geography (-122 [-388 , $+83$]), though still slightly below in Rio (-231 [-338 , -125]). The mixture does not recover the idealized specialist’s advantage ($+2020$ against $+2713$ there).

5.5 Robustness in the two-disease variant

Two earlier analyses, run without retraining in the two-disease variant on the Rio surfaces, complement the cross-geography study. Across epidemic years (dengue 2015 with chikungunya 2016, and Rio 2016 alone), results are practically unchanged (C: $+2692$, $+2710$, $+2721$; $C \times A$ between $+2009$ and $+2077$, $P \geq 0.98$), as expected from surfaces that are all about 54% informative. Across the whole range of spatial informativeness obtained by sharpening, from 0.50 to 0.94, the advantage of per-case credit holds ($C \times A$ between $+2000$ and $+2350$, P between 0.89 and 1.00; Figure 9). C’s reward stays flat: an agent trained where location is uninformative does not exploit it when it becomes informative—consistent with the idealized-geography result, where only an agent *trained* there learns to use the map.

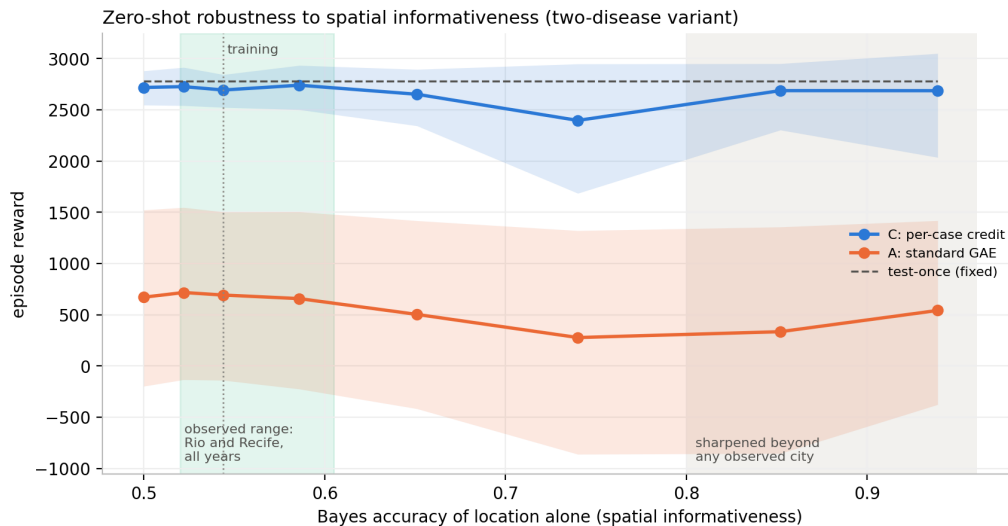


Figure 9: Zero-shot evaluation as spatial informativeness varies (two-disease variant; 95% intervals over training seeds and outbreaks). The green band is the range observed in Rio de Janeiro and Recife; the gray band requires sharpening beyond any observed city.

6 Discussion

The obstacle to learning a testing policy was not where it first seemed. For much of this project, “the agent does not test” was treated as a problem of exploration, reward scale or algorithm. Part of it was: a stabilized DQN reliably learns inaction, and PPO was the first learned agent to beat doing nothing. But the decisive bottleneck was credit assignment. When decisions about many patients interleave in epidemic time, the standard return carries almost no information about whether a test was useful; the return along the patient’s own timeline carries almost all of it. Estimating advantages per case, in days, recovers that signal without touching the environment or the reward, and the result holds in every geography we tested.

What the agent learns depends on the problem it faces. With only two diseases, one test suffices and the agent learns *whether* to investigate. With non-arboviral suspects, investigation has depth, and the agent learns a policy as good as sequential testing guided by the clinician—the best rule we could write—while ordering 8% fewer tests. Where location is informative, it learns to use the map and surpasses every rule with far fewer tests. Matching an expert rule where the rule is near-optimal, and beating it where there is exploitable structure, is the pattern one would hope for from a policy learned from reward alone. A useful by-product is itself practical: among fixed rules, testing first for the disease the clinician suspects is substantially better than always testing dengue first (+350 in reward and 73 fewer tests per episode).

The transfer study turns the weak spatial signal of real data into a design lesson. Because location carries little information in real cities, the best policies there are spatially invariant: the Rio-trained agent ignores its spatial inputs and, for that reason, transfers to another city. Policies that do rely on space are brittle—the idealized-geography agent because the signal disappears in real cities, the Recife agent because it learned a cue tied to one urban footprint. Training on real geography that fills the observation, or on several geographies with enough training, avoids the worst of this; a principled way to make spatial inputs comparable across cities of different size and shape remains open.

6.1 Limitations

- **Training seeds.** Three training seeds per arm and geography give the upper level of the hierarchical bootstrap little resolution; intervals are honest about outbreaks and optimistic about training runs.
- **Spatial scale.** The environment grid is 400×400 for every city, so its physical scale differs between cities (the 20-cell radius of epidemiological confirmation is about 1 km in Recife and 3.5 km in Rio), and the Rio surface extends over the sea while the Recife surface is masked to the municipality. The Recife transfer failure is tied to these differences and is not resolved.
- **Simulated clinicians and tests.** Clinician accuracy is symmetric between diseases and drawn per episode; test parameters are fixed; non-arboviral suspects are uniform in space. Recife’s discarded notifications could provide an empirical surface, but a notification discarded for dengue may be chikungunya.
- **Data.** Rio surfaces are built from notified cases and Recife surfaces from confirmed cases geocoded to street or neighborhood level. In 2015–2016 Recife was the epicenter of the Zika epidemic, and part of the dengue confirmed by clinical–epidemiological criteria in those years is likely Zika.
- **Externality of tests.** Per-case credit does not credit a test for its effect on future epidemiological confirmations nearby; the agents barely use epidemiological confirmation.

- **Capacity and seasonality.** Scarcity is modeled through test cost, not a daily laboratory capacity; the SEIR model has no seasonal forcing.

6.2 Future work

The most immediate steps are a per-case analysis of which cases the agent tests twice, which would turn “the agent chooses which cases need a second test” into a measured statement; a physically consistent spatial scale across cities, which would let the Recife transfer failure be tested properly; and additional years of Rio de Janeiro notifications as held-out tests. Beyond that: synthetic populations from the 2022 Brazilian census [30] to build scenarios for other cities with realistic population density; a daily laboratory capacity, which would turn scarcity into explicit prioritization; and a credit assignment that accounts for the externality of positive tests.

7 Conclusion

We framed the allocation of laboratory tests in arbovirus surveillance as a sequential decision in epidemic time, in a simulator grounded in surveillance data from two Brazilian cities. In real data, the location of a case barely reveals whether it is dengue or chikungunya, so tests are the main source of information—and standard deep RL fails to learn to use them, because the value of a test is lost among interleaved decisions about other patients. Estimating advantages along each patient’s timeline, in epidemic days, makes the agent investigate in every geography we tested. In real cities it matches the best expert rule with fewer tests and carries that policy to held-out epidemics of another city; where location is informative, it beats the rule outright. Which rule is best depends on the case mix, and whether a learned policy transfers depends on whether it leans on spatial cues that the next city does not share.

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A Work in progress (internal; remove before submission)

- **[TODO: Per-case analysis of which cases C tests twice (first result, clinical label, evidence about the clinician).]**
- **[TODO: Physically consistent spatial scale across cities (radius and grid in km; mask Rio to the municipality), then retrain Recife and re-test the transfer.]**
- **[TODO: Held-out years of Rio de Janeiro (data requested from Fiocruz) as additional test geographies.]**

- [TODO: More training seeds (at least 5) for the headline comparisons.]
- [TODO: Related work: RL for testing and epidemic control; multi-agent credit assignment; shortcut learning and domain randomization; test allocation in surveillance.]
- [TODO: Anchor RT-PCR parameters in references; redo the test-cost sweep in the full environment.]
- [TODO: Check flagged bibliography entries (marked in `refs.bib`).]